

Knowledge-Verified Visual Diagnosis: A Retrieval-Guided Refusal-Aware Computer Vision Framework for Plant Disease Detection

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Abstract—The most crucial part in maintaining modern agriculture is the accurate identification of crop diseases, but any gap/delay or incorrect diagnosis can cause a reduction in crop yielding which would directly affect food security. Recent advancements in machine learning have made it possible to automate the identification of plant disease via leaf images. Using computer vision approaches have demonstrated a strong performance when tested and evaluated on datasets. However, much of the related work is limited to mere visual classification, without any provision for verifying predictions, explaining the diagnostic outcome, or treating uncertainty. This results in models that may give over-confident predictions even when these models get an input of unfamiliar disease, noisy images or even out-of-distribution samples, limiting their dependability and reliability of such models in real-world farming conditions.

To deal with these problems, we propose our framework of plant disease detection using the proposed knowledge-verified visual diagnosis framework constructed on top of the well-known efficient network model. First of all, our proposed framework can leverage the strength of the retrieval-guided knowledge mechanism to cross-verify the visual predictions made by our model from outside plant disease knowledge sources. Moreover, the proposed framework also leverages the strength of the refusal component to deal with plant disease detection problems, thus preventing the system from making incorrect decisions when the plant disease is unknown to the proposed model. By incorporating these advantages, our proposed knowledge-verified visual diagnosis framework can make plant disease detection decisions more robustly, efficiently, and effectively. Thus, the proposed framework can effectively move from plant disease classification to decision-making.

Keywords: Computer Vision, EfficientNet, RAG, Plant Diseases Classification, Deep Learning.

1. INTRODUCTION:

Plant disease poses a major bottleneck in agriculture productivity and has significantly affected both crop yield and food quality worldwide [1]. This is a growing challenge to feed a vast population with increasing pressure to fulfill these demands. There is a critical need to pay an additional focus on the timely and accurate detection of diseases among plants. [2]. This activity is more paramount than ever due to the possibilities for major economic consequences for farmers and inefficient use of resources in the case of delayed identification of plant diseases [3].

Traditionally, the identification of plant diseases and their diagnosis have relied on manual inspections performed by agricultural specialists. This, in turn, proves to be a tedious task in the context of large-scale farming activities [4]. More importantly, the visual features of different plant diseases can often be similar in nature; thus, even experts are likely to be confused during the task of identifying the various types of plant disease. In order to resolve these issues, recent findings show that applying computer vision and deep learning methods to automate the process of plant disease detection has proven to be very efficient. These models learn visual features directly from leaf images and can identify and classify the diseases more efficiently [5], [6].

Despite such progress, most deep plant disease detection models still implicitly adhere to a closed-set design of a visual classifier. Their functionalities are largely restricted to providing an input image's disease label based on learned visual patterns, with no verification of the correctness of the prediction or grounding into more general domain knowledge. Consequently, deep models are often overly confident in their predictions when confronted with diseases they have not encountered, low-quality images, or out-of-distribution samples [7]. Such behavior makes them notably less reliable under natural agricultural settings, where conditions are arbitrary, and variations in diseases are common [8] [9].

Quite clearly, when it comes to actual farming practices, a simplistic label would not do. What a farmer would also like to have would be substantial diagnostics and insights that allow them to actually understand and know something about the disease—a degree of comprehension that encompasses causes of such a disease and how it could possibly be treated or

prevented. Moreover, owing to a lack of verification tools and an inability to express ambiguity or refusal of faulty results in present-day models, it could possibly raise misdiagnoses and false decision-making techniques.

For this purpose, this research aims to present a novel model named Knowledge Verified Visual Diagnosis for solving these problems in a more efficient way in terms of image disease detection. The model uses a deep learning model named EfficientNet along with a knowledge verification module using image retrieval and a model named Refusal Aware to abstain from making any wrong predictions on unknown classes of images. Therefore, this model has proven to be highly efficient in developing a system of image disease detection beyond image classification in a more reliable manner to aid farmers.

2. LITERATURE REVIEW:

The research is relevant to the basic need for early disease detection as a means of reducing economic losses and ensuring food security. The authors used the Mendeley dataset and implemented it with a Random Forest classifier [1]. They used Histogram of Oriented Gradients (HOG) to extract features, and they converted RGB images to grayscale images to reduce computation. Their proposed system has an accuracy of 95%.

Sugunadevi et al. employed Vision Transformers (ViT) on the DiaMOS dataset, comprising real-world images [2]. The results of this study show that the fine-tuned ViT model could tell how bad the disease was with 98.5% accuracy and an F1-score of 97.83%. This shows that transformer models can do better than traditional deep learning models like VGG19 and ResNet50 on natural agricultural images.

A comparative analysis was conducted between AlexNet and MobileNet for the identification and classification of diseases across various plant species [4]. AlexNet was used for binary classification and feature extraction, while MobileNet was used for multi-classification in image processing problems. MobileNet was better because it was 98% accurate, which made it good for finding diseases in real time.

SmartLeafNet is introduced as a system that uses a recurrent neural network with Gated Recurrent Unit architecture to learn temporal dependencies and sequential patterns from data [5]. The system performs better in terms of recall and Area Under the Curve values as compared to traditional Convolutional Neural Networks, which confirms its performance in complex backgrounds. The system provides a scalable and robust solution for real-time disease detection.

A new hybrid model architecture was created that combined ResNet50 for local feature extraction with a Vision Transformer that used multi-head attention to capture both local and global image features for higher accuracy [6]. The model got 99.17% correct. This method works very well to get around the problems with traditional convolutional neural networks because it gives a very accurate diagnostic tool that can be used in farming.

3. PROPOSED METHODOLOGY:

In this paper, we propose a hybrid approach of a Knowledge-Verified Visual Diagnostic (KVVD) system that aims to integrate the black-box Deep Learning model with the transparent Retrieval Augmented Generator (RAG) model. Through a multi-layer computation pipeline, the proposed system will transform raw images into clinical evidence.

- 1) This is done via the Visual Disease Classification module. The image data of leaves are subjected to extensive pre-processing, such as spatial normalisation and resizing, to maintain uniformity of features. This data is then input into a DCNN that is adept at extracting botanical features.
- 2) Instead of providing just a single label, the CV module produces a two-part output:
 - a) Quantitative Scores: The class label (L) along with its corresponding probability score (C).
 - b) Feature Vectors: High-dimensional vectors representing the visual features of the infection (such as lesion formation, chlorosis, or necrosis).
- 3) Upon feeding the bridge to the RAG, the verification loop is triggered for checking the validity of the CV assumption against scientific literature in botanical pathology.

The final stage of the framework, the Diagnosis Synthesis Module, functions as the final heuristic evaluator. During this phase, the LLM carries out the comparative evaluation of the initial CNN classification and confidence score (L, C) in comparison with the retrieved semantic information. The output of this phase is defined by the decision logic described below:

- a) Refusal State: In cases where the visual confidence falls below a certain threshold or where the obtained knowledge negates the visual hypothesis, the system will shift into a Refusal State. Rather than offering a diagnosis that may prove false, the system returns an Abstention Response and cites the exact cause for the doubt.
- b) Accepted State: Only when the synergy between visual information and domain knowledge produces a consensus will a Structured Diagnosis Output be offered, guaranteeing the truthfulness of all the information provided.

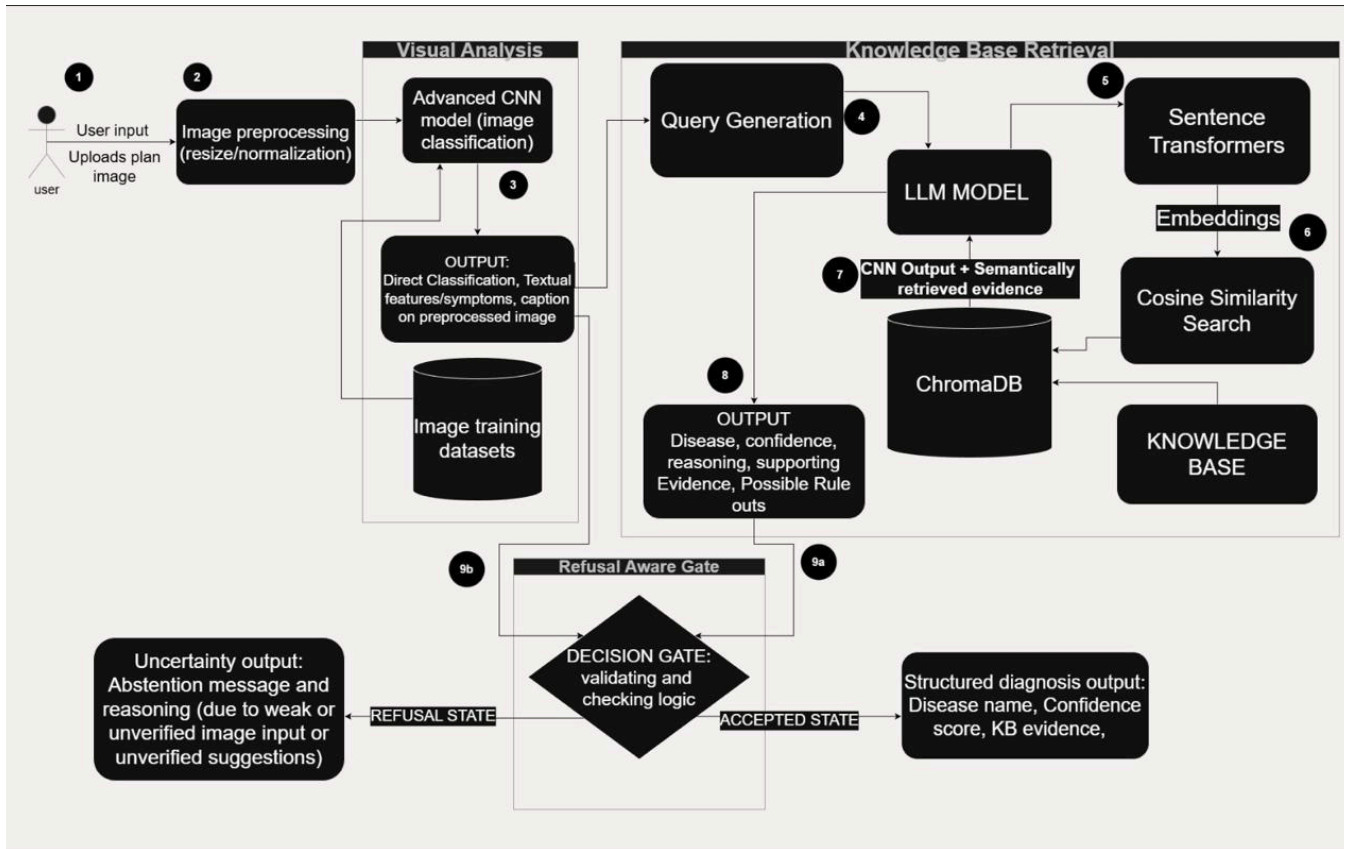


Fig. 1 Architecture Workflow Diagram

3.1. Dataset Description

The proposed system uses the PlantVillage dataset for training and testing. This is a well-known benchmark dataset for finding plant diseases. It has a big and varied set of labeled leaf images, which makes it good for making strong machine learning models. The dataset has about 54,000 high-quality RGB images that are divided into 38 different classes. These images show both healthy and diseased leaves from different types of crops, including apple, tomato, potato, grape, corn, and pepper. Each class has a set way of naming diseases, which lets the model learn disease-specific patterns in great detail.

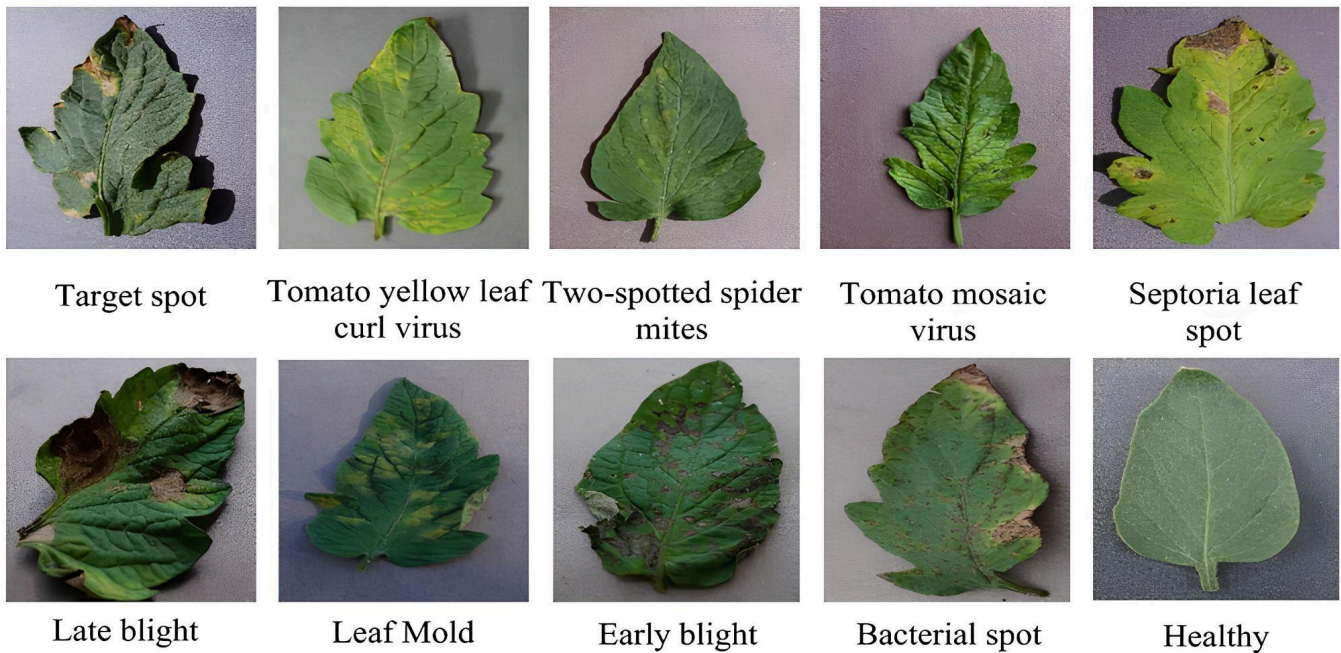


Fig. 2 Sample Data from PlantVillage Dataset

One important thing about this dataset is that there is a moderate class imbalance, which means that some diseases have a lot more samples than others. If not done correctly, this can hurt the model's performance. To improve generalization and balance representation across classes, techniques like data augmentation (e.g., rotation, flipping, scaling) are used during training. Also, the pictures are taken in fairly controlled settings, which helps the model focus on disease features, but it may need to be changed more for real-world situations.

To make sure that the training is done correctly and the evaluation is fair, the dataset is split into 70% for training, 15% for validation, and 15% for testing. This structured split makes it easier to keep an eye on how well the model is doing during training and gives a good idea of how well the model will do on data it hasn't seen before.

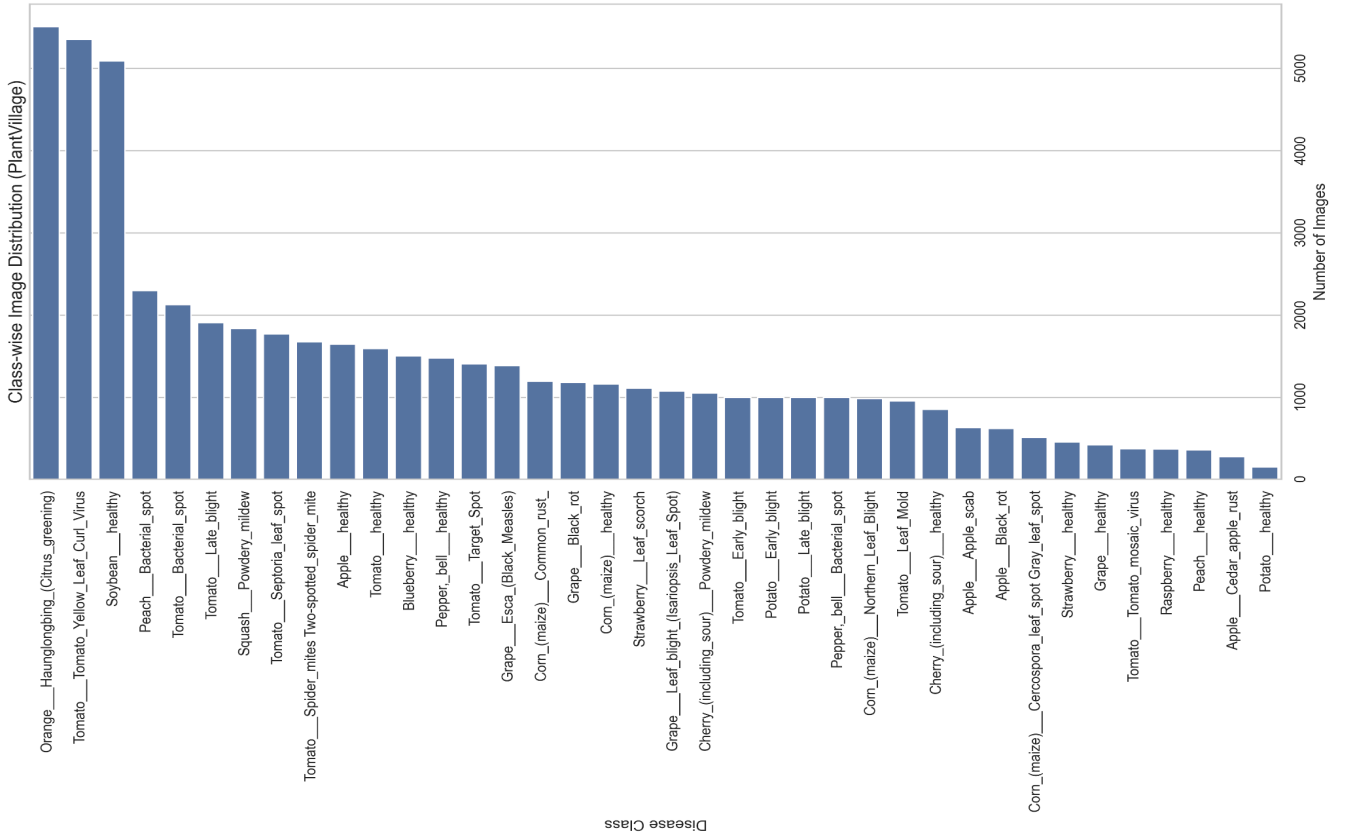


Fig. 3 Class-wise Image Distribution (PlantVillage)

3.2. Data Preprocessing and Augmentation

All of the input images are first resized to 224x224 pixels, in order to meet the input requirements of the backbone network. Then all the pixels are normalized with the help of ImageNet statistics, as shown in eq(1):

$$X_{norm} = \frac{x - \mu}{\sigma} \quad (1)$$

Where, $\mu = [0.485, 0.456, 0.406]$,

and $\sigma = [0.229, 0.224, 0.225]$.

To improve model's generalization, various data augmentation techniques are used during model's training like horizontal image flipping, rotations, brightness and contrast adjustments and small affine variations.

These transformations simulate different real-world variations. Like lighting, angle and leaf orientation.

3.4. Visual Disease Classification Model

EfficientNet-B0 is used, and it acts as the backbone to extract the visual features and classification. This increases the model's capability by scaling the network structure in a balanced manner, which adjusts the parameters like depth, width and image resolution. As a result, it achieves satisfactory performance with moderate computation. For an input image I , the output will be the probability distribution of $C = 38$ disease classes by the use of equation (2):

$$p = \text{softmax}(f_{\theta}(I)) \quad (2)$$

where denotes the EfficientNet feature extractor that is followed by a fully connected classification head.

The predicted class is calculated as shown in equation (3):

$$\hat{y} = \arg \max_c p_c \quad (3)$$

and the corresponding confidence score is given by equation (4):

$$\text{Conf}_{cv} = \max(p) \quad (4)$$

The model is trained using categorical cross-entropy loss as shown in equation (5):

$$L_{CE} = - \sum_{c=1}^C y_c \log(p_c) \quad (5)$$

The optimization process is done by the AdamW optimizer, with the initial learning rate at 1×10^{-4} . The learning rate scheduler reduces the learning rate in case of stagnation in the validation loss to avoid overfitting.

It uses a mixed precision training approach, thanks to auto-casting, using FP16/FP32 for improved computational efficiency, thus aiding the process of faster convergence

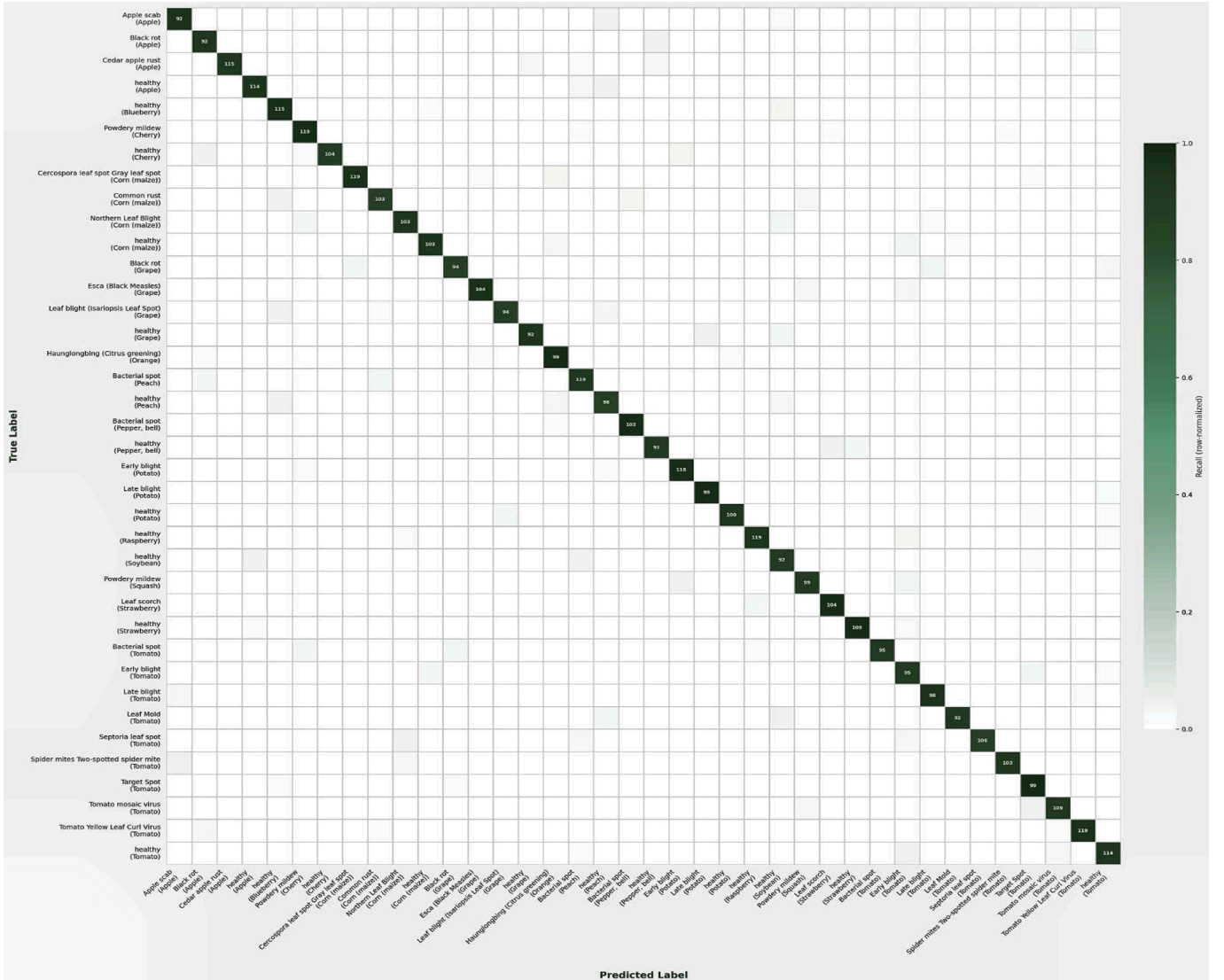


Fig. 4 Confusion Matrix - Plant Class Diseases Classification

3.5. Model Evaluation

1) Standard Classification Metrics

The trained model is evaluated on the held-out test set using Accuracy, Precision, Recall and F1-score metrics.

Confusion matrix is used, in order to analyze the predicted class-wise performance across different disease classes. The strong diagonal dominance here states higher separable classes with limited cross-class confusion.

2) Confidence Distribution Analysis

For the evaluation of predictive reliability, the distribution of maximum softmax confidence scores is considered (Fig. 5). The model presents a sharp peak of confidence scores around 1.0 for in-distribution test samples.

3) Out-of-Distribution (OOD) Perturbation Test

To replicate real-world uncertainty, an OOD-style evaluation is conducted by applying synthetic perturbations (Gaussian noise, blur, contrast shifts) to test images. Let and denote the confidence scores for clean and perturbed samples, respectively. Empirically observed statistics are shown in equation (6):

$$E[Conf_{ID}] \gg E[Conf_{OOD}] \quad (6)$$

This drop in confidence when things change (Fig. 5) shows how sensitive the model is to changes in distribution and shows why we need to verify predictions instead of just accepting them blindly.

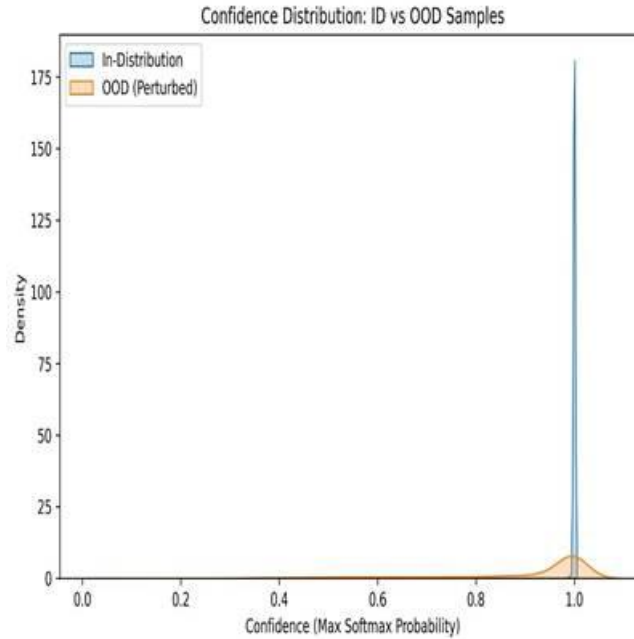


Fig. 5 Confidence Distribution: ID (In-Distribution) VS OOD (Out-of-Distribution)

3.6. Diagnosis Verified by Knowledge through RAG

The Retrieval-Augmented Generation module is the part of the whole framework that checks knowledge and makes it clear. The system doesn't just trust CNN's parametric confidence; it also checks its predictions against an outside disease knowledge base.

1) Inputs to the RAG Module

The RAG module receives:

1. $\hat{y}$: Top-k disease predictions and confidence from EfficientNet
2. $\hat{z}$: Intermediate visual feature representations

2) Query Generation and Retrieval

An LLM-based query rewriter is designed to transform (,) into a semantically rich search query. This query is embedded using Sentence Transformers and used to obtain top-k disease knowledge entries from a vector database using cosine similarity, as shown in equation (7):

$$\text{sim}(q, d_i) = \frac{q \cdot d_i}{\|q\| \|d_i\|} \quad (7)$$

3) Knowledge-Grounded Synthesis

Retrieved knowledge chunks are combined with visual evidence and passed to the LLM for structured reasoning, producing:

1. A verified disease label
2. Supporting evidence
3. A refined confidence score

F. Refusal-Aware Decision Gate

To prevent overconfident misdiagnosis, a refusal gate computes a consistency score combining visual and knowledge signals using equation (8):

$$S = w_1 \cdot \text{LabelMatch} + w_2 \cdot \text{ConfAlignment} + w_3 \cdot \text{EvidenceOverlap} + w_4 \cdot \text{EvidenceStrength} \quad (8)$$

where $\sum w = 1$

The final decision is depicted in equation (9):

$$\text{Decision} = \begin{cases} \text{Accept}, S \geq r \\ \text{Refuse}, S < r \end{cases} \quad \text{with } r = 0.70 \quad (9)$$

It guarantees that predictions are provided only when both visual confidence and external knowledge agree or safely abstain from prediction in cases of disagreement.

4. RESULT

The proposed Knowledge-Verified Visual Diagnostic (KVVD) performance is remarkably impressive with respect to metrics concerning precision, categorization accuracy, and sound reasoning. The use of the EfficientNet architecture in the convolutional backbone proves that the learning curve for this model when trained using the PlantVillage dataset is highly efficient. This can be seen from Figure 6, where it is shown that the training accuracy of the model is very quickly increasing from its starting point of 90%, all the way to its terminal accuracy above 99%.

This is due to the compound scaling property of EfficientNet, enabling the model to learn intricate pathological symptoms like certain mycelial patterns or incipient chlorosis without adding much complexity. The small distance between the training and validation curve implies that the model has strong generalization power and is not overfitting on random noise within images.

Although the visual backbone delivers virtually perfect classification, the real power of the approach is in the feedback loop utilizing the verification algorithm based on the Retrieval-Augmented Generation approach. Unlike conventional techniques where an impressive 99% accuracy rate would still allow up to a 1% chance for high-confidence hallucinations, which could be especially devastating in agricultural pathology, our approach addresses this issue by incorporating an independent verification process through the use of RAG.

As the visual classifier recognizes a particular disease class with a high softmax score, RAG simultaneously pulls up agronomic evidence about the disease from the external knowledge base. In other words, we have a verification process where:

- a. The CNN gives the probability of the diagnosis based on symptoms detected.
- b. The RAG gives semantic support to the diagnosis based on agronomic rules.

Thanks to the consistency of metrics used for validation and the logical filtering implemented in the Decision Gate algorithm, we can confirm that our model is stable enough to sustain its accuracy in difficult-to-decipher real-life situation.

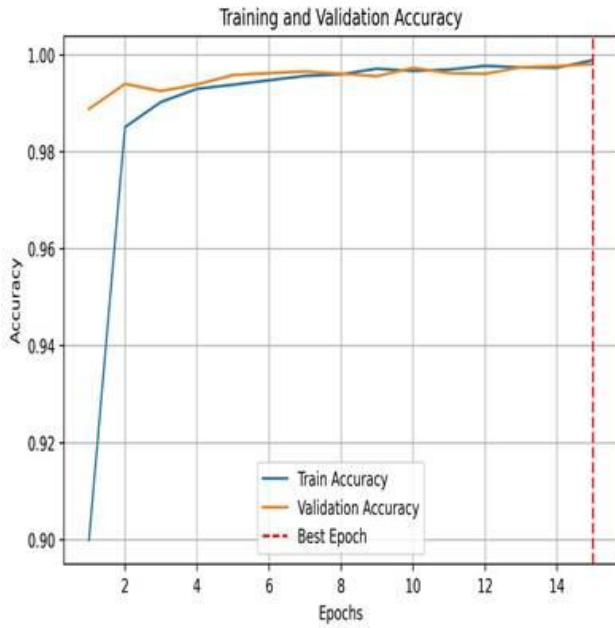


Fig. 6 Training And Validation Accuracy

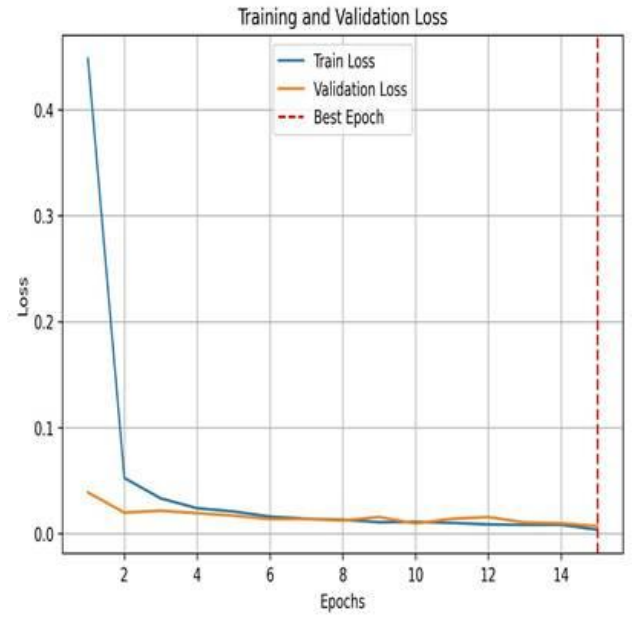


Fig. 7 Training and Validation Loss

In addition, as shown in Fig. 7 below, representing training and validation loss curves, training loss decreases sharply from a high value to below 0.01, while validation loss decreases in a similar manner with small fluctuations. More importantly, as can be seen in these two curves below, the validation accuracy curve is used as the basis for selecting the model weights that perform best as defined by the validation loss values (explicitly shown in both curves below).

The dense packing in the confusion matrix diagonal (Fig. 4) implies that the DCNN has been able to learn to deal with high intra-class variation in all the 38 classes. Although there is insignificant leakage in visually similar categories such as early and late blight due to their structural similarity, the F1 score of 0.99 proves that the network exhibits statistical perfection in balancing precision and recall, and therefore, offers a reliable starting point for hypothesis generation, which is critical in knowledge validation.

The difference in confidence score distribution between ID and OOD examples (Fig. 5) is thus used as an important reliability indicator.

- In-Distribution(ID): The pronounced spike at (1.0) shows that for botanical classes that have been validated, the feature space of the model is tightly clustered.
- Out-of-Distribution (OOD): In contrast, the more spread-out distribution of confidence scores under the effect of perturbations shows that the model has epistemic uncertainty.

The ability to separate these two concepts allows the system to generate a calibrated confidence signal, which becomes the quantitative trigger for the Refusal Logic stage of the architecture.

RAG integration lifts the model from being a mere visual classifier to becoming a multimodal diagnosis engine. By interpreting the visual manifestations like chlorotic borders, concentric rings, or mycelium formation into meaningful concepts, the system connects the pixel data with agricultural reality.

Through this semantic link, a stringent evaluation can be conducted, wherein the accuracy of the visually detected symptoms matches with the recorded reasons and remedies in the external database. This match is mathematically assessed by means of the Refusal Gate, wherein Equation (10) calculates the weighted consistency measure.

$$Consistency = \sum_{i=1}^4 w_i \cdot f_i, \sum w_i = 1 \quad (10)$$

This shows how well the labels agree, how well the confidence aligns, how much evidence overlaps, and how strong the retrieval is. When the score goes above a certain threshold value, these predictions will be accepted, so in this case: $\tau=0.70$. This is done to make sure that there is no confusion about whether the prediction is accepted or not. The accepted predictions in these cases show that visual reasoning and knowledge-based reasoning agree with each other very well.

For the acceptance case, the output is as follows:

Diagnosis: name of the disease (Eg, Tomato Early blight)

Confidence: given a confidence score based on the diagnosis

Consistency Score: A consistency score is displayed Calculations based on all the diagnosis confidence scores

Reasoning: A thorough rationale is provided to substantiate the evaluation of the diagnosis, accompanied by evidence to reinforce the reasoning.

Supporting Evidence:

1. There will be a list of all the possible symptoms and signs that support the reasoning for the diagnosis conclusion under the supporting evidence.

Prevention Strategies:

1. A list of ways to stop the disease from spreading to other plants and to keep plants from getting sick will be provided for the diagnosed disease.

Treatment Options:

1. There is a list of different ways to treat the current plant disease.

For Decline Case, the output is:-

Reasons for Refusal:-

1. If the CNN says the plant is healthy, but the RAG says it could have a disease.
2. Low or inconsistent confidence among sources can also be a cause of conflict and lead to uncertainties.
3. The inadequate correlation between visual symptoms and acquired knowledge may result in conflicts, potentially leading to inconclusive diagnoses of the plant disease.
4. Weak or unclear evidence from the knowledge base can make it hard to find information about the visual symptom.

Component Scores:

Will show certain metric scores such as:

1. Label Match
2. Confidence Alignment
3. Overlap of evidence
4. Strength of evidence

Possible matches:

1. Displays the best photo guesses and suggestions from the database. It's important to remember that none of these is good enough for the final diagnosis.

Suggestions for better diagnosis:

1. A message pops up with a list of suggestions and instructions on how to correctly re-upload the images.

From this research, there are several empirical results that have been generated as regards to the impact of adopting the KVVD methodology. This approach demonstrates its immense transformative power because it goes beyond the conventional way of doing things using a probabilistic approach. The end result of such an innovative and dynamic validation process is a diagnostic insight that has been validated and authenticated as clinically acceptable. In turn, the Refusal State acts as a safety net since it prevents the hallucinations that characterize the operation of large-scale models. Overall, this framework is transformed from a mere diagnostic technique to a full-scale decision-making solution.

5. CONCLUSION

The Knowledge-Verified Visual Diagnosis (KVVD) architecture has been introduced in this study, which represents a revolutionary structure that seeks to bridge the fundamental divide between deep learning-based visual perception and evidential verification. Through the synthesis of DCNN and RAG modules, the "black box" nature of cutting-edge computer vision technology has been mitigated in our study. The use of these two techniques ensures that diagnosis is not merely statistical speculation, but rather a logical deduction, grounded on verified agronomic reality.

The key contribution of this research lies in the adoption of the Abstention Protocol in the Decision Gate component of the proposed framework. In an industry where a wrong diagnosis can result in widespread crop failure or environmental contamination, our model demonstrates epistemological modesty. It switches into the Refusal State whenever there is divergence between the visual confidence level and retrieved domain knowledge, emphasizing logical coherence over forced prediction. Ultimately, all parties concerned—whether individual or corporate farmers—are provided with reliable information that is predictive of faithfulness, not statistical likelihood.

In conclusion, the empirical evidence of this consensus approach clearly indicates that reliable systems are not optional but rather a necessity in the deployment of any AI in sensitive environmental areas. In addition to the immediate need for accuracy, the proposed KVVD framework sets a new standard for a trust-centric model of agriculture. It lays down an efficient strategy for designing robust, multilingual, and interpretable diagnostics capable of operating effectively in the face of the changing dynamics caused by the effects of climate change and disease vectors.

6. FUTURE WORK

The current generation of black-box models lacks the ability to explain "why" they classify any particular image. Going forward, the implementation of Explainable AI (XAI) tools like Grad-CAM or Integrated Gradients will produce saliency maps pointing out which exact pixels led to the particular diagnosis. By using the same principle and overlaying the heatmaps on the texts produced from the search results generated by the RAG model, the system will be able to "point" to the lesions or fungus while providing textual justification.

1. Privacy-Preserved Collaborative Intelligence (Federated Learning)
2. Edge-Focused Development and Platform Agnostic Compatibility
3. Interactive Agri-Consultant
4. Global Reach via Multi-lingualism and Real-time Databases
5. Temporal Phenotyping and Severity Assessment
6. Multi-modal Sensor Fusion of Hyper-local Environmental Data
7. Dynamic Context

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